ENGR 1050
Homework 4

Due: Monday, November 2, 2026 by 11:59 PM

Instructions

We've now been using our ODE solver for a variety of problems for a couple weeks. Also in lab we've been building up toward designing a helicopter that can hover in place. In this homework, we'll help you fine-tune the control policy for the helicopter, which we saw in the class doesn't do a perfect job.

This week's assignment will be primarily *mathematical*. Submit photos/scans of your derivation along with a Jupyter notebook for any code that you write. Assignments must be submitted individually.

Recall that our proportional control policy is given by:

$$\dot{\theta} = \omega$$

$$\dot{\omega} = -b\omega - K\sin(\theta) + T$$

where b is a damping coefficient, K is a constant related to gravity and the length of the helicopter arm, and T is the torque applied by the helicopter's fan. The control policy we are using for T is:

$$T = K_p(\theta_{\text{target}} - \theta)$$

where T is the torque applied by the helicopter's fan, K_p is a proportional gain that we can tune, θ_{target} is the desired angle of the helicopter, and θ is the current angle of the helicopter.

Problems

Problem 1: Steady state behavior of policy (50 points)

We will analyze the steady state behavior of our policy. To make the math simpler, replace $\sin(\theta)$ with θ (this is a reasonable approximation for small angles).

Make the following assumptions about steady state behavior:

- $\dot{\theta} = 0$ (the helicopter is not moving)
- $\dot{\omega} = 0$ (the angular velocity is not changing)
- $\theta = \theta_{\text{ss}}$ (the helicopter is at some steady state angle)

- $\omega = 0$ (the angular velocity is zero)

Derive an equation for the steady state angle θ_{ss} as a function of K_p, K, b, and θ_{target}. Show all work.

Problem 2: Stability analysis (50 points)

Use your analysis to consider two scenarios:

- **Scenario 1.** You set $\theta_{\text{target}} = \pi/4$ (45 degrees), and use your calibrated values for K and b from the in-class exercise. Use the formula you derived in Problem 1 to explain whether it is possible to achieve $\theta_{\text{ss}} = \pi/4$ for any value of K_p. Confirm your analysis by running the code from the in-class exercise.

- **Scenario 2.** You are free to set θ_{target} to any value of your choice (potentially different from $\pi/4$). Provide choices of K_p and θ_{target} that give the desired steady state $\theta_{\text{ss}} = \pi/4$. Confirm your analysis by running the code from the in-class exercise.